A.I Fundamentals

Module 4 Study Guide

AI in Financial Services

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: Banking, Payments, and Customer Service

In this lesson: AI Onboarding & Support; Instant Caribbean Payments; Cost Breakdown Exampl

Key points

- AI speeds up onboarding and support @ "transforms onboarding through instant digital identity checks"
- Smart routing lowers transaction processing costs @ "lowers payment fees through smart network routing"
- Digital tools preserve monthly income and time @ "saves you hundreds of dollars in fees and lost time every single month"

Examples from the lesson

- Today, we are kicking off Module 4, AI in Financial Services, with our very first lesson, Banking, Payments, and Customer Service.
- Have you noticed how opening a bank account or getting customer support feels different today than it did five years ago?
- Instead of waiting on hold for twenty minutes just to check your account balance, confirm a transfer, or flag a lost debit card, an AI assistant answers instantly on a 24/7 basis.
- Imagine visualising your monthly financial management, starting with a base salary of TT$ 6,000.
- Under traditional banking, taking time off work and traveling to a physical branch costs you roughly TT$ 150 in lost wages and transit fees.
- You also face manual account processing fees of TT$ 50 and wire processing costs of TT$ 100, leaving you with TT$ 5,700 in effective monthly value.
- Now, let's look at that same TT$ 6,000 monthly scenario when using an AI-optimized digital banking setup.
- Digital onboarding eliminates lost work hours entirely, dropping your transit and lost wage cost down to TT$ 0.
- Automated identity verification lowers your account setup fee to TT$ 10, while smart transaction routing brings your transfer fees down to just TT$ 15.
- That brings your total digital friction cost down to just TT$ 25.
- When you compare these two pathways, the financial benefit becomes crystal clear.

- Subtracting your digital setup costs of TT$ 25 from the traditional manual friction cost of TT$ 300 gives you a direct monthly savings of TT$ 275.
- Over the course of a full year, that simple shift to AI-powered banking keeps an extra TT$ 3,300 right inside your local credit union or savings account.

Lesson 2: Fraud Detection and Credit Decisions

In this lesson: AI Pattern Recognition in Fraud; Automated Credit and Risk Scoring; Worked Ex

Key points

- AI flags unusual foreign transactions in milliseconds
- Automated scoring evaluates monthly cash flow for loan approvals
- Consumers hold rights to transparency, explanations, and human review

Examples from the lesson

- We are currently in Module 4, "AI in Financial Services," and today’s focus is Lesson 2: "Fraud Detection and Credit Decisions."
- Imagine you are resting at home in San Fernando on a Friday night, but suddenly your credit card is charged TT$3,500 for a luxury item in an online store overseas.
- The AI recognizes that this purchase breaks your pattern in three distinct ways: the unusual hour, the foreign merchant, and the large dollar amount.
- Suppose Marcus earns a monthly income of TT$8,000 as a technician and applies for a small loan to repair his maxi taxi.
- His rent is TT$2,500, his utility bills total TT$500, and his existing credit union loan payment is TT$1,000.
- This brings his total monthly fixed expenses to TT$4,000.
- Subtracting TT$4,000 in expenses from his TT$8,000 income leaves Marcus with TT$4,000 in remaining monthly cash flow.
- The automated system then evaluates the requested loan payment of TT$800 per month against this available balance.
- Since the new payment takes up just 20 percent of his remaining cash flow, the algorithm generates an instant pre-approval recommendation for the credit officer.
- If an algorithm declines your application because of an unusual expense pattern, a human credit officer can review your full context, such as a one-time family emergency, and adjust the outcome.
- Artificial intelligence protects your accounts by spotting unusual spending patterns in real time, like a sudden TT$3,500 foreign charge while you are safe at home.
- It speeds up lending decisions by calculating your true cash flow, such as approving Marcus when his TT$800 loan payment comfortably fits his TT$4,000 monthly surplus.
- In our next lesson, Lesson 3: "Personal Finance and Wealth Tools," we will explore how AI-powered tools can help you build personal budgets, set savings goals, and grow your wealth right here in Trinidad and Tobago.

Lesson 3: Personal Finance and Wealth Tools

In this lesson: Evaluate digital budgeting apps; Understand robo-advisor algorithms; Apply fou

Key points

- Evaluate apps using fees, transparency, privacy, and oversight
- Analyze complete TT$ budget allocations for automated accounts
- Maintain human control while leveraging AI wealth tools

Examples from the lesson

- We are in Module 4, AI in Financial Services, and this is Lesson 3: Personal Finance and Wealth Tools.
- However, to use these technologies safely and effectively, you must evaluate them against four critical criteria: transaction fees, algorithm transparency, data privacy, and human oversight.
- Imagine you earn a net monthly salary of TT$ 10,000 and want to automate your finances responsibly using AI personal finance tools.
- Your rent or mortgage payment takes TT$ 3,500, utilities and groceries take TT$ 2,500, and transport including maxi fares and fuel costs TT$ 1,000.
- That brings your total fixed expenses to TT$ 7,000 per month.
- You set aside TT$ 1,000 for discretionary spending like dining out or Carnival savings, and you route TT$ 2,000 into an automated robo-advisor investment account.
- Your total allocations equal TT$ 3,000, leaving you with a net monthly cash balance of TT$ 0.
- Before letting this automated plan run, you audit the digital tools using our four criteria.
- The robo-advisor charges an annual management fee of 0.5% on your TT$ 2,000 monthly contributions, which equals TT$ 10 per month.
- Today, we learned how to evaluate AI personal finance tools, robo-advisors, and budgeting assistants using four key criteria: fees, transparency, data privacy, and human oversight.
- We also analyzed a practical TT$ 10,000 monthly allocation to ensure automated systems align with disciplined personal habits.

Recap checklist — can you say each one?

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Lesson 3: Personal Finance and Wealth Tools

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☐ Analyze complete TT$ budget allocations for automated accounts

☐ Maintain human control while leveraging AI wealth tools